

CARL ALCOTT

Systems & Network Administrator | IT Infrastructure & Security Engineer

Lake of the Ozarks, MO • carl@arcfob.space • github.com/OzarkMountainPirate

PROFESSIONAL SUMMARY

Systems and Network Administrator with 20 years of hands-on experience architecting, securing, and operating complex IT infrastructure across managed-services and K-12 environments. Deep expertise in enterprise networking (VLAN segmentation, routing, wireless, multi-vendor firewalls), Windows and Linux systems administration, virtualization, and security operations — SIEM, IDS/IPS, EDR/MDR, and vulnerability management. Proven record leading full-stack projects from design through deployment, hardening, and documentation, with an emphasis on defense-in-depth security, and automation.

CORE COMPETENCIES

Security Operations: SIEM (Elastic/ELK, AlienVault, Huntress), IDS/IPS (Suricata, CrowdSec), EDR/MDR, vulnerability scanning (OpenVAS), incident response, log analysis, security awareness training (SAT), CIPA compliance, email security (SPF/DKIM/DMARC)

Networking: LAN/WAN, VLAN segmentation, inter-VLAN routing, VPN, enterprise wireless, multi-vendor switching & firewalls, campus area networks (CAN), TCP/IP, DNS, DHCP

Systems Administration: Windows Server, Active Directory, Group Policy, Linux (Debian/Ubuntu), Exchange / Exchange Online, Microsoft 365, Entra ID / Azure AD, Google Workspace for Education

Endpoint & Device Management: JAMF / JAMF Pro, Apple macOS / iOS / iPadOS, Chromebook & Google Admin console, one-to-one device fleets (200-300 devices per site), imaging, enrollment & deployment

Virtualization & Storage: VMware ESXi / vCenter (vMotion, HA), Hyper-V, Docker, SAN/NAS, iSCSI, NFS, SMB/CIFS, ZFS, backup & disaster recovery

Automation & IaC: PowerShell, Bash, Python, Ansible, Terraform, Git / Gitea, declarative configuration management

Voice & UC: 3CX (Advanced Certified), Avaya, VoIP provisioning, telephony / paging and building-system integrations

PROFESSIONAL EXPERIENCE

Independent Infrastructure & Security Engineering

2025 – Present

Self-Directed Research & Development — Lake of the Ozarks, MO

- Design, build, and operate a multi-host Linux lab to develop and validate production-grade security and infrastructure tooling, applying enterprise security-operations practices end to end.
- Built a full detection pipeline: Suricata IDS (EVE) → syslog-ng → Logstash → Elasticsearch → Kibana, with GeoIP enrichment and custom Kibana dashboards for network threat visibility.
- Deployed CrowdSec intrusion prevention across a three-host fleet with per-host reverse-proxy bouncers for automated, behavior-based blocking.
- Engineered a secure edge and remote-access stack using Cloudflare Tunnels, sslh, OpenVPN (tls-crypt), and Traefik reverse proxy.
- Hardened Linux hosts with nftables, fail2ban, auditd, LUKS full-disk encryption, ZFS snapshots/replication, SSH hardening, and least-privilege Docker (socket-proxy, file-based secrets).
- Manage all configuration as version-controlled declarative infrastructure (Git/Gitea) and publish selected tooling as open source (see Technical Projects).

Chief Technology Officer / Senior Systems Engineer

2016 – 2025

KCHelpdesk (Managed Services Provider) — Independence, MO

- Served as senior technical lead and CTO for an MSP whose core client base was private schools, churches/parishes, alongside a broad commercial portfolio spanning healthcare, dental, veterinary, architecture, manufacturing, and production, banking, and municipal government.
- Designed, deployed, and administered campus area networks for K-12 environments — VLAN segmentation, inter-VLAN routing, enterprise wireless, and multi-vendor switching from server room to endpoint.
- Deployed and managed multi-vendor firewalls and content-filtering / CIPA-compliance systems, from proxy-based platforms (DansGuardian, SquidGuard) to modern DNS based filtering (Lightspeed, GoGuardian).
- Administered student information systems including PowerSchool and Infinite Campus, and managed Google Workspace for Education across one-to-one device fleets.

- Deployed and integrated 3CX VoIP telephony in school and business environments, including integrations with intercom/paging, elevator, fire-alarm, and security-panel systems.
- Migrated clients to Microsoft 365 and Exchange Online; managed email security, spam filtering, and domain authentication (SPF, DKIM, DMARC).
- Architected and maintained VMware ESXi / vCenter (vMotion, HA) and Hyper-V environments; implemented on-prem and cloud backup and disaster-recovery solutions.
- Administered Windows Server (AD, DNS, DHCP, Group Policy) and Linux infrastructure; managed SAN/NAS storage with iSCSI/NFS/SMB and replication; automated tasks with PowerShell and Bash.
- Implemented and maintained security-operations tooling — EDR/MDR (Huntress), SIEM (AlienVault), vulnerability scanning (OpenVAS), and security awareness training; delivered Tier 3 escalation support and long-term IT strategy for business continuity.

Director of Technology / Network & Systems Administrator

2014 – 2016

Wentworth Military Academy & College — Lexington, MO

- Directed all campus technology for a residential military academy and college, engineering a high-density network for ~1,100 concurrent devices (heavy student BYOD) atop a ~500-endpoint managed fleet.
- Administered Windows Server infrastructure, Active Directory, O365, Blackbaud, Moodle, plus a campus-wide Avaya VoIP telephony system.
- Led VMware-based virtualization projects and campus-wide infrastructure upgrades.
- Oversaw backups, patch management, disaster recovery, and account security; trained staff and documented processes for continuity and audit readiness.

Systems / Network Administrator

2011 – 2014

KCHelpdesk — Independence, MO

- Supported Active Directory, Group Policy, Exchange, and Hyper-V virtual machines across small to mid-sized business environments.
- Assisted with server upgrades, patching, and deployment of client projects; managed DNS, DHCP, and networking.

System / Network Administrator & Shop Manager

2009 – 2011

Lexington Tech Solutions — Lexington, MO

- Provided full-stack IT support including servers, networking, and client systems; managed customer repair operations and small-business IT contracts.

Support / Helpdesk Technician

2006 – 2009

Kearney School District — Kearney, MO

- Delivered district-wide helpdesk and infrastructure support, supporting servers, desktops, printers, and classroom technology across multiple buildings.
- Assisted with district-wide system upgrades and end-user training and support.

Field Technician (contract)

2005 – 2006

S&A Computers — Midwest and Arizona (travelling)

- Branch-office hardware refresh for Bank of America as a subcontractor to HP, across sites in the Midwest and Arizona.

TECHNICAL PROJECTS & OPEN SOURCE

- **suricata-elk** — IDS-to-SIEM pipeline (Suricata → syslog-ng → Logstash → Elasticsearch → Kibana) with GeolIP enrichment. github.com/OzarkMountainPirate/suricata-elk
- **utilities** — Collection of Linux system-administration and automation tooling. github.com/OzarkMountainPirate/utilities
- **ark-fleet** — A fun learning project for testing and practicing **DevOps methodology and tooling** against a real workload. <https://github.com/OzarkMountainPirate/ark-fleet>

EDUCATION

- CIS Security / Information Assurance Coursework — DeVry University, 2008–2010
- Diploma, Computer Technology — Vatterott College, 2004–2005
- Certificate, Computer Maintenance & Networking — Lex-La-Ray Technical Center, 2002–2004

PROFESSIONAL INVOLVEMENT

- Advisory Committee Member — Lex-La-Ray Technical Center (2019–Present)
- IT Demonstrator — Missouri Vocational Educators Conference (2017–2019)